INT302 Image Processing – Lab 2 Report

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Lab Title: Eigenface-Based Face Recognition and Histogram Equalization

1 Introduction

This lab implements an eigenface-based face recognition system and a histogram equalization image enhancement method. The eigenface part is based on Principal Component Analysis (PCA), where face images are represented by a small number of projection weights instead of raw pixels. The main tasks include checking whether the provided eigenfaces form an orthogonal and orthonormal basis, projecting face images onto the eigenface space, reconstructing faces from weights, recognizing employees using Euclidean distance, and testing the robustness of the algorithm under noise and luminance changes. Finally, histogram equalization is implemented manually to enhance a low dynamic range image without using the built-in `histeq` function.

The purpose of this lab is not only to obtain recognition results, but also to understand why eigenfaces can compress high-dimensional face images into a low-dimensional feature vector, and where this method is likely to fail in practical situations such as strong impulse noise or over-exposure.

2 Orthonormal Basis (10 marks)

2.1 Verification of Orthogonality (5 marks)

2.1.1 Method

Each eigenface is originally stored as a two-dimensional image of size 450×300. To check orthogonality, each eigenface is first reshaped into a column vector. If the eigenfaces are orthogonal, the inner product between any two different eigenfaces should be close to zero:

$$\langle e_i, e_j \rangle = e_i^T e_j \approx 0, \quad i \neq j$$

Therefore, I compute the dot product matrix:

$$D = E^T E$$

where each column of E is one vectorised eigenface. The diagonal elements represent the squared norm of each eigenface, while the off-diagonal elements represent the inner products between different eigenfaces.

2.1.2 MATLAB Code

```
[M, N, K] = size(eigenfaces_blk);
pixels = M * N;

ef_2d = reshape(eigenfaces_blk, pixels, K);

D = ef_2d' * ef_2d;
D_off = D - diag(diag(D));

max_off_diag = max(abs(D_off(:)));

fprintf('Eigenface dimensions: %d x %d pixels\n', M, N);
fprintf('Number of eigenfaces: %d\n', K);
```

```
fprintf('Maximum absolute off-diagonal value: %.4e\n', max_off_diag);

figure;
imagesc(D);
colorbar;
title('Dot Product Matrix of 100 Eigenfaces');
xlabel('Eigenface Index');
ylabel('Eigenface Index');
```

Listing 1: Orthogonality Verification Code

2.1.3 Result

The eigenface dimensions are 450×300 pixels, and there are 100 eigenfaces in total. The maximum absolute off-diagonal value is:

$$\text{max off-diagonal value} = 1.4268 \times 10^{-7}$$

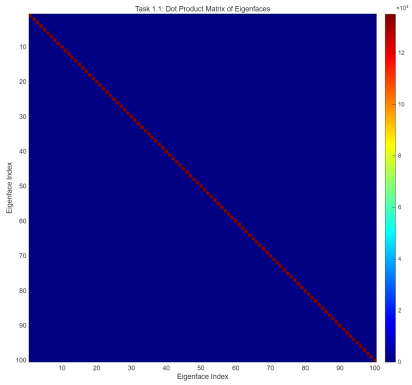


Figure 1: Dot Product Matrix of 100 Eigenfaces

Analysis and Conclusion The dot product matrix is almost diagonal, which means that different eigenfaces have almost zero inner product. The small non-zero value 1.4268×10^{-7} is very close to zero and is mainly caused by floating-point numerical precision in MATLAB. In theory, PCA produces eigenvectors that are mutually orthogonal because they correspond to different principal directions of the covariance matrix. Therefore, the result confirms that the provided eigenfaces are orthogonal enough for projection and recognition.

2.2 Orthonormality Check and Normalization (5 marks)

2.2.1 Method

Orthogonality only requires different basis vectors to have zero inner product. Orthonormality additionally requires each basis vector to have unit length:

$$\|e_i\|_2 = 1$$

Therefore, I calculate the L2 norm of each eigenface. If the norm is not 1, each eigenface is normalized as:

$$\hat{e}_i = \frac{e_i}{\|e_i\|_2}$$

2.2.2 MATLAB Code

```
norms_before = sqrt(sum(ef_2d.^2, 1));

fprintf('Norms of original eigenfaces:\n');
fprintf('Min: %.6f\n', min(norms_before));
fprintf('Max: %.6f\n', max(norms_before));
fprintf('Mean: %.6f\n', mean(norms_before));

ef_2d_norm = ef_2d ./ norms_before;

norms_after = sqrt(sum(ef_2d_norm.^2, 1));

fprintf('Norms after normalization:\n');
fprintf('Min: %.6f\n', min(norms_after));
fprintf('Max: %.6f\n', max(norms_after));
fprintf('Mean: %.6f\n', mean(norms_after));

eigenfaces_blk_norm = reshape(ef_2d_norm, M, N, K);
```

Listing 2: Orthonormality Check and Normalization

2.2.3 Result

Norms of original eigenfaces:

- Minimum norm: 367.423461
- Maximum norm: 367.423461
- Mean norm: 367.423461

Norms after L2 normalization:

- Minimum norm: 1.000000
- Maximum norm: 1.000000
- Mean norm: 1.000000

Analysis The original eigenfaces are orthogonal but not orthonormal because their norms are 367.423461 instead of 1. After normalization, all eigenfaces have unit norm while their directions remain unchanged. This is important because projection weights should represent the contribution of each eigenface direction, not the scale of the basis vector itself. If the eigenfaces were not normalized, the magnitude of the weights could be affected by the scaling of the eigenfaces, which would make distance-based recognition less reliable.

3 Evaluating Eigenface Weights (20 marks)

3.1 Function: `get_face_weights` (10 marks)

3.1.1 Method

To represent a face using eigenfaces, the face image is first vectorised. The image vector is then projected onto each eigenface. The projection coefficient is the weight of that eigenface:

$$w_i = e_i^T x$$

where x is the vectorised input face and e_i is the i-th normalized eigenface. For all eigenfaces, the weight vector is:

$$w = E^T x$$

Each weight measures how strongly the corresponding eigenface pattern appears in the input image.

3.1.2 MATLAB Function

```
function [weights_of_face] = get_face_weights(im, eigenfaces_blk)

    [M, N, K] = size(eigenfaces_blk);

    im = double(im);
    im_vec = im(:);

    ef_2d = reshape(eigenfaces_blk, M*N, K);

    weights_of_face = ef_2d' * im_vec;
end
```

Listing 3: get_face_weights Function

3.2 Compute and Plot Weights for `find_id.jpg` (10 marks)

3.2.1 Result

- Input image size: 450×300
- Eigenface size: 450×300
- Number of eigenfaces: 100
- Size of weight vector: 100×1

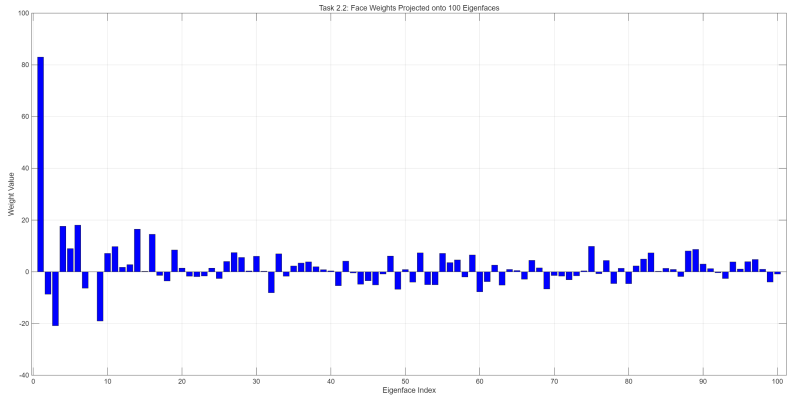


Figure 2: Eigenface Weights of `find_id.jpg`

Analysis The plot does carry useful information about the employee's face, but not in the same way as the original image. Instead of showing pixels directly, the plot shows how much each eigenface contributes to the face representation. Large positive or negative weights indicate that the corresponding eigenface pattern is important for describing this person, while weights close to zero indicate little contribution.

The first few weights have larger magnitudes, which is expected because the early PCA components usually capture the most significant variations in the face dataset, such as general facial structure, lighting direction and broad shape. Later components have smaller weights and usually describe finer details. Therefore, the weight vector can be treated as a compact facial signature. Two images of the same employee should have similar weight vectors, while different employees should have larger distances between their weight vectors.

4 Face Generation from Weights (20 marks)

4.1 Function: `generate_face_from_weights` (10 marks)

4.1.1 Method

A face can be reconstructed by taking a weighted linear combination of all eigenfaces:

$$\hat{x} = Ew = \sum_{i=1}^{K} w_i e_i$$

where w_i is the weight of the i-th eigenface and $K = 100$ in this lab. The reconstructed vector is then reshaped back into a 450×300 image.

4.1.2 MATLAB Function

```
function [im] = generate_face_from_weights(weights_of_face, eigenfaces_blk)

    [M, N, K] = size(eigenfaces_blk);

    ef_2d = reshape(eigenfaces_blk, M*N, K);

    im_vec = ef_2d * weights_of_face(:);

    im = reshape(im_vec, M, N);

    im = im - min(im(:));
    im = im / max(im(:));
end
```

Listing 4: generate_face_from_weights Function

4.2 Reconstructed Face (3 marks)



Figure 3: Original and Reconstructed Face of `find_id.jpg`

Analysis The reconstructed image preserves the main identity-related facial structure, including the overall face shape, hair region, eyes, nose and mouth position. However, compared with the original image, the reconstructed image is smoother and slightly blurred. This is because only 100 eigenfaces are used, so high-frequency details and small texture variations cannot be fully recovered. This is an acceptable result for recognition because the goal is not perfect visual reconstruction, but preservation of discriminative facial information.

A limitation is that reconstruction quality depends on whether the input face is similar to the training face distribution. If the input image has very different pose, expression, illumination or background, the same 100-dimensional eigenface space may not reconstruct it well.

Theoretical Insight This function implements a linear mapping from a low-dimensional representation back to the high-dimensional image space. The eigenfaces span a subspace often referred to as the "face space", and the weight vector represents the coordinates of the input face within this subspace.

The reconstruction process can therefore be interpreted as a linear transformation that maps these low-dimensional coordinates back into the original pixel space. In other words, the reconstructed image is obtained by combining the eigenfaces according to their corresponding weights.

Since PCA preserves the directions of maximum variance in the dataset, even a relatively small number of eigenfaces is sufficient to retain the dominant structural information of the face. As a result, the reconstructed image is able to preserve the overall facial geometry and identity characteristics, even though some fine details are inevitably lost.

4.3 Why 100 Parameters Are Enough (7 marks)

Analysis The original face image contains $450 \times 300 = 135000$ pixels, yet only 100 parameters are sufficient to achieve a high-quality reconstruction. This can be explained by the following fundamental reasons.

Reason 1: Intrinsic low-dimensional manifold structure of face images

Although the pixel space is extremely high-dimensional, natural face images do not occupy this entire space. Instead, they lie on a much lower-dimensional manifold. This is because all human faces share a common topological structure, such as the relative positions of eyes, nose and mouth. Variations between faces (identity, expression, illumination and pose) are constrained and therefore span only a small subset of all possible pixel configurations. PCA effectively captures this low-dimensional manifold by identifying the principal directions of variation.

Reason 2: Strong redundancy and correlation between pixels

Neighbouring pixels in face images are highly correlated. For example, pixel values in smooth regions such as skin tend to change gradually, and adjacent pixels often have very similar intensities. PCA exploits this redundancy by transforming correlated pixel values into a set of uncorrelated principal components. Each principal component represents a global variation pattern rather than individual pixel values.

By retaining only the first 100 principal components, PCA preserves the directions with the largest variance, which correspond to meaningful facial structures and identity information. The discarded components usually represent noise or very fine details with low variance. Therefore, even though the dimensionality is drastically reduced, the essential facial information is still preserved, allowing accurate reconstruction.

5 Face Recognition (25 marks)

5.1 Function: `get_employee_ID_from_DB` (13 marks)

5.1.1 Method

To recognize a face, the input image is projected into the eigenface space to obtain its weight vector. Then this weight vector is compared with every employee weight vector stored in `employees_DB`. The closest employee is selected using Euclidean distance:

$$d(w, w_j) = \sqrt{\sum_{i=1}^{K} (w_i - w_{j,i})^2}$$

where w is the input face weight vector and w_j is the stored weight vector of the j-th employee. The employee with the smallest distance is returned as the recognition result.

5.1.2 MATLAB Function

```
function [ID] = get_employee_ID_from_DB(im, employees_DB, eigenfaces_blk)

    input_weights = get_face_weights(im, eigenfaces_blk);
    input_weights = input_weights(:)';

    num_employees = length(employees_DB);

    min_distance = inf;
    ID = -1;

    for i = 1:num_employees

        db_weights = employees_DB(i).weights;
        db_weights = db_weights(:)';

        distance = sqrt(sum((input_weights - db_weights).^2));

        if distance < min_distance
            min_distance = distance;
            ID = employees_DB(i).id;
        end
    end

    fprintf('Minimum distance found: %.4f\n', min_distance);
    fprintf('Recognized Employee ID: %d\n', ID);
end
```

Listing 5: get_employee_ID_from_DB Function

Analysis This function follows the nearest-neighbour classification principle. Euclidean distance is suitable here because each face has already been represented as a vector in the same eigenface feature space. A smaller Euclidean distance means that the two faces have more similar PCA coefficients and therefore are more likely to belong to the same person.

Compared with raw pixel matching, this method is more efficient because it compares 100-dimensional weight vectors instead of 135000-dimensional image vectors. It also reduces the influence of minor pixel-level variations because PCA keeps the dominant face structure rather than every local detail.

5.2 Recognition Result for `find_id.jpg` (2 marks)

5.2.1 Result

The recognition result is:

$$\text{Recognized Employee ID} = 81$$

The minimum distance found is:

$$33.4200$$

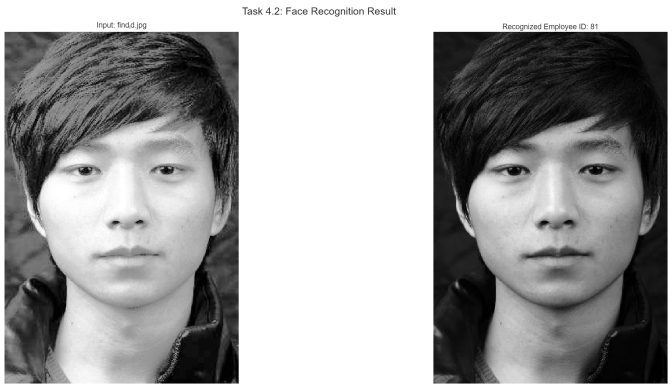


Figure 4: Recognition Result for `find_id.jpg`

Analysis The system correctly recognizes `find_id.jpg` as employee ID 81. This result shows that the weight vector of the input image is closest to the stored weight vector of employee 81 in the database. Since the database is a closed-set database, the system always assigns the input image to the nearest stored employee. Therefore, the recognition result is reliable only if the person is actually included in the database.

5.3 Test with Personal ID Image (10 marks)

5.3.1 Procedure and Result

A personal ID photo was used to test whether a new face can be added into the database. The procedure is:

1. Convert the personal photo to grayscale.
2. Resize it to 450×300 to match the eigenface dimensions.
3. Normalize the intensity range.
4. Compute the eigenface weights of the personal photo.
5. Add the new ID and weight vector into `employees_DB`.

6. Run the recognition function again.

The result is:

- New personal ID: 111
- Recognition distance: 0.4481
- Recognized ID: 111
- Recognition result: successful



Figure 5: Preprocessed Personal Photo

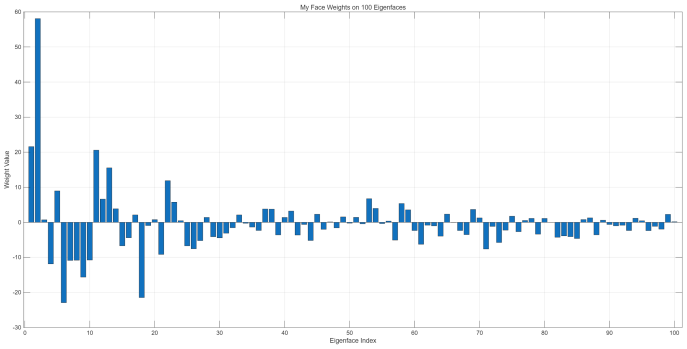


Figure 6: Eigenface Weights of Personal Photo

Analysis The personal image is recognized correctly because the same image, after preprocessing and projection, is compared against its own stored weight vector in the database. Therefore, the distance is very small. This confirms that the eigenface framework can add a new employee by storing only the new weight vector, without recalculating PCA eigenfaces.

However, this test is easier than a real recognition scenario because the enrollment image and test image are highly similar. In practice, a more convincing test would use two different photos of the same person under different lighting, pose or expression. If the result becomes worse, possible improvements include using multiple enrollment images per person, applying illumination normalization, cropping the face more accurately, and using a more robust distance metric such as cosine distance.

6 Algorithm Robustness (30 marks)

The following tests use `find_id.jpg` as the input image. The correct employee ID is 81.

6.1 Robustness to AWGN (7 marks)

σ	Min Distance	Recognized [Y/N]
5	32.9466	Y
20	31.5298	Y
60	22.0447	Y
100	10.2674	Y
150	9.4366	Y

Table 1: Recognition under AWGN

Analysis The algorithm correctly recognizes the employee under all tested AWGN levels from $\sigma = 5$ to $\sigma = 150$. This shows that the eigenface method is relatively robust to additive Gaussian noise. One reason is that AWGN is random and distributed across the whole image, while eigenfaces represent dominant global facial patterns. During projection, random noise components that are not aligned with the main eigenface directions have limited influence on the final weight vector.

An interesting observation is that the minimum distance decreases as the noise level increases. This does not necessarily mean that the image becomes easier to recognize. A more likely explanation is that the noise and normalization process change the overall scale of the projected weights. Even though the absolute distance changes, the nearest employee remains ID 81, so the relative ranking is preserved. This suggests that nearest-neighbour recognition depends more on relative distances than on the absolute distance value alone.

Root Cause Analysis The fundamental reason is related to the dimensionality of the PCA feature space. When only a very small number of eigenfaces is used, for example $K = 5$, the low-dimensional PCA space loses part of its ability to suppress noise through redundancy. PCA can reduce Gaussian noise effectively only when enough principal components are retained, because the noise energy can be distributed across many dimensions and weakly projected onto the main face subspace. However, when K is too small, the feature dimension is much lower than the intrinsic dimensionality required for reliable face discrimination. In this case, the noise energy is forced into only a few retained principal components, so the first few eigenface weights may be seriously contaminated.

As a result, the weight vector of the same person under noisy conditions may drift far away from its clean database representation. In extreme cases, the distance between noisy images of the same person may become larger than the distance to another employee. This means the feature drift crosses the class boundary in the PCA space. Therefore, a very small K value weakens the natural filtering effect of PCA and makes the algorithm much more sensitive to AWGN.

6.2 Robustness to Salt and Pepper Noise (7 marks)

Noise Density	Min Distance	Recognized [Y/N]
1%	32.2666	Y
20%	15.3421	Y
40%	6.9996	Y
60%	21.9176	N, ID 33
80%	36.0638	N, ID 64

Table 2: Recognition under Salt and Pepper Noise

Analysis The system remains correct up to 40% salt and pepper noise, but fails at 60% and 80%. Salt and pepper noise is more destructive than AWGN because it replaces pixels with extreme black or white values. At low density, only isolated pixels are corrupted, so the global face structure is still visible. However, when

the density becomes very high, many important facial regions are replaced by impulse noise. This changes the projected weight vector significantly and causes the nearest-neighbour result to shift to other employees.

This result shows a limitation of PCA-based recognition: PCA is good at representing global linear variations, but it is not designed to reject extreme local outliers. A median filter would be more suitable before recognition because it can remove impulse noise while preserving edges better than simple averaging.

Root Cause Analysis The main reason for failure under salt and pepper noise is that a small PCA feature space does not provide a robust encoding mechanism for abnormal pixels. Salt and pepper noise is an extreme pixel-level corruption, where some pixels are directly replaced by black or white values. During PCA projection, each eigenface weight is calculated as a linear weighted sum of all image pixels. When K is small, each retained eigenface carries a large amount of information, and some local corrupted pixels can have a disproportionately large influence on the final weight vector.

In a higher-dimensional PCA space, the disturbance caused by corrupted pixels can be spread across more principal components, so the influence on each individual component is relatively smaller. However, with a very small K, the same disturbance is concentrated into only a few weights. This amplifies the local damage caused by impulse noise and produces unstable and unpredictable feature changes. Therefore, the recognition system may collapse when the density of salt and pepper noise becomes high.

6.3 Robustness to Luminance Change (10 marks)

6.3.1 Brighter Image, $\rho > 1$

ρ	Min Distance	Recognized [Y/N]
1.001	33.0585	Y
1.010	34.2192	Y
1.100	38.7599	N, ID 70
1.200	41.4249	N, ID 70
1.300	41.7258	N, ID 70

Table 3: Recognition under Brighter Luminance

6.3.2 Darker Image, $\rho < 1$

ρ	Min Distance	Recognized [Y/N]
0.99	32.3531	Y
0.90	24.4972	Y
0.85	20.6905	Y
0.80	15.9716	Y
0.70	8.5600	Y

Table 4: Recognition under Darker Luminance

Analysis The result shows an obvious asymmetry between brightening and darkening. The system is robust to darkening down to $\rho = 0.7$, but it fails when the image is brightened to $\rho = 1.1$ or above.

For darker images, the pixel intensities are mainly scaled down. Although the image becomes darker, many relative contrast relationships between facial regions are still preserved. Therefore, the eigenface weights still remain close enough to the correct employee.

For brighter images, the situation is different. When brightness is increased too much, some pixels become saturated and clipped to the maximum intensity. Once clipping happens, the lost facial contrast cannot be recovered by simple scaling. Important regions such as skin texture, nose shadow and face boundary may become flattened. This changes the PCA projection and leads to wrong recognition. The repeated wrong

ID 70 also suggests that over-brightened images move the input weight vector toward a different region of the feature space.

Root Cause Analysis The fundamental reason for sensitivity to luminance variation is that a small PCA feature space cannot clearly separate illumination variation from identity variation. PCA ranks principal components according to variance. In face images, global brightness changes often produce large variance, so the first few eigenfaces may encode illumination-related patterns rather than purely identity-related patterns. When only a small number of eigenfaces is used, such as $K = 5$, most available feature dimensions may be occupied by brightness information, leaving insufficient dimensions for illumination-invariant identity features.

This explains why luminance change can cause unstable recognition. If the brightness change moves the input image along a direction covered by the retained eigenfaces, the system may still match correctly. However, if the luminance change moves the image towards the boundary or outside of the limited PCA subspace, the distances to database samples become unreliable. In this case, the classifier may choose an incorrect nearest neighbour. To resist luminance changes more effectively, more eigenfaces should be retained so that the model can represent both illumination variation and identity-related information in separate directions.

6.4 Possible Solutions to Improve Robustness (6 marks)

Based on the above experiments, the following methods can improve robustness:

1. **Median filtering:** This is suitable for salt and pepper noise because it removes isolated black and white impulse pixels while preserving facial edges.
2. **Gaussian smoothing:** This can reduce AWGN by averaging local random variations, although excessive smoothing may also remove useful facial details.
3. **Histogram equalization:** This can normalize contrast and make the image less sensitive to low dynamic range or uneven lighting.
4. **Illumination normalization:** Methods such as gamma correction or homomorphic filtering can reduce the effect of lighting changes before PCA projection.
5. **Data augmentation:** Storing multiple versions of each employee under different lighting and noise conditions would make the nearest-neighbour database more representative.
6. **Alternative distance metrics:** Cosine distance can be used as an additional comparison method because it focuses on the direction of the eigenface weight vector rather than its absolute magnitude. Normalized Euclidean distance can also make matching less affected by global brightness scaling.

Comparison between Euclidean Distance and Cosine Distance In the current recognition function, Euclidean distance is used to compare the input weight vector with each employee's stored weight vector:

$$d_E(w, w_j) = \sqrt{\sum_{i=1}^{K} (w_i - w_{j,i})^2}$$

This metric measures the absolute distance between two weight vectors. It is simple and effective when all face images have similar illumination and contrast. However, when the input image becomes brighter or darker, the magnitude of the weight vector may change even if the identity-related pattern is still similar. In this case, Euclidean distance may become larger or smaller mainly because of scale changes rather than true identity differences.

Cosine distance provides a useful comparison because it measures the angle between two weight vectors:

$$d_C(w, w_j) = 1 - \frac{w \cdot w_j}{\|w\|_2 \|w_j\|_2}$$

A smaller cosine distance means that the two vectors point in a more similar direction. Therefore, cosine distance is less sensitive to global magnitude changes in the eigenface weights. This is especially useful for luminance variation, because brightness changes may scale the projection weights while preserving part of the relative facial pattern. For this reason, cosine distance could improve robustness under moderate illumination changes.

However, cosine distance is not always better than Euclidean distance. Since it ignores vector magnitude, it may lose useful information when the magnitude of certain PCA coefficients is important for distinguishing different employees. It also cannot fully solve severe over-brightness, because saturation and clipping destroy facial texture information rather than only changing the vector scale. Therefore, a practical improvement would be to compare both Euclidean distance and cosine distance, or to use cosine distance after illumination normalization and weight normalization.

Summary Overall, the eigenface recognition system is strongest against AWGN, moderately robust against low and medium salt and pepper noise, and weak against over-brightness. The main reason is that PCA preserves global face structure but cannot fully handle severe outliers or saturated illumination. Therefore, preprocessing is necessary if the system is used in a real access-control environment.

7 Histogram Equalization (15 marks)

7.1 Algorithm Description

Histogram equalization is a spatial-domain contrast enhancement method. It redistributes pixel intensities so that the output image uses a wider dynamic range. For a grayscale image with intensity levels in $[0, 1]$, the main steps are:

1. Count the number of pixels at each intensity level to obtain the histogram.
2. Normalize the histogram to estimate the probability distribution:

$$p(r_k) = \frac{n_k}{MN}$$

   where n_k is the number of pixels at intensity level r_k.

3. Compute the cumulative distribution function:

$$CDF(r_k) = \sum_{j=0}^{k} p(r_j)$$

4. Map each original intensity value to a new value:

$$s_k = CDF(r_k)$$

The CDF is a monotonically increasing function, so the order of brightness values is preserved. Dark pixels remain relatively dark and bright pixels remain relatively bright, but the contrast is stretched.

7.2 MATLAB Function

```
function img_eq = my_histogram_equalization(img)

    img = double(img);

    if max(img(:)) > 1
        img = img / 255;
    end
```

```
    L = 256;

    [counts, ~] = histcounts(img(:), L, 'BinLimits', [0, 1]);

    pdf = counts / numel(img);

    cdf = cumsum(pdf);

    bin_idx = ceil(img(:) * L);
    bin_idx(bin_idx < 1) = 1;
    bin_idx(bin_idx > L) = L;

    img_eq = cdf(bin_idx);

    img_eq = reshape(img_eq, size(img));
end
```

Listing 6: my_histogram_equalization Function

7.3 Result and Findings

- The original image is `lenna512_low_dynamic_range.bmp`.
- The original histogram is concentrated in a narrow middle intensity range, which explains the low contrast.
- After histogram equalization, the intensity values are spread over a wider range.
- The enhanced image shows clearer texture and stronger contrast, especially in regions that were originally flat or unclear.
- The output histogram is not perfectly uniform because the image is discrete and many pixels may share the same intensity value.

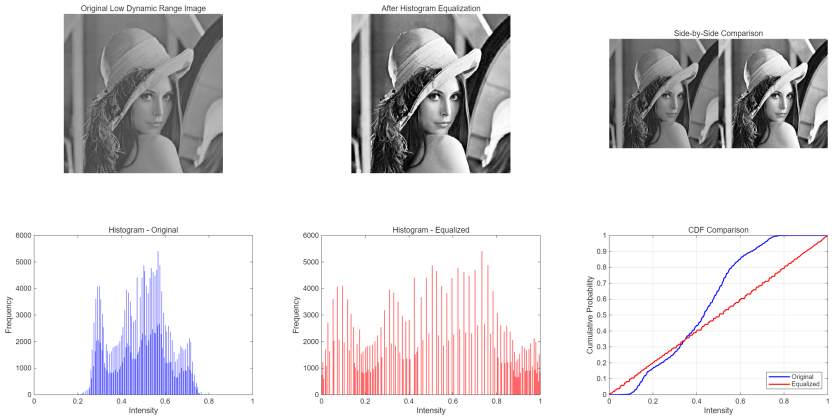


Figure 7: Histogram Equalization Results

Analysis The result confirms that histogram equalization successfully improves the contrast of the low dynamic range image. The original histogram is compressed into a limited intensity interval, so many visual details are difficult to observe. After equalization, the CDF mapping stretches this narrow range into a wider

range. This increases the difference between neighbouring intensity regions and makes the image visually clearer.

However, histogram equalization also has limitations. It enhances global contrast but does not consider local image content. Therefore, it may over-enhance noise or produce unnatural contrast if the original image already contains strong local variations. For face recognition, histogram equalization can help reduce lighting differences, but it should be used carefully because excessive contrast changes may also alter the eigenface weights.

8 Overall Conclusion

In this lab, the eigenfaces were verified to be orthogonal and were normalized to form an orthonormal basis. The face image `find_id.jpg` was successfully projected into a 100-dimensional eigenface weight vector and reconstructed from these weights. The reconstructed image preserved the main facial identity information, although some high-frequency details were lost.

The recognition system correctly identified `find_id.jpg` as employee ID 81 using Euclidean nearest-neighbour matching in eigenface space. A personal image was also added into the database and recognized successfully, showing that new employees can be enrolled by storing their weight vectors.

The robustness experiments showed that the system is reliable under AWGN, acceptable under moderate salt and pepper noise, but sensitive to strong impulse noise and over-bright luminance changes. This indicates that eigenface recognition is effective but should be combined with preprocessing methods such as median filtering, illumination normalization and histogram equalization in practical applications.

Finally, histogram equalization was implemented manually and successfully enhanced the contrast of a low dynamic range image. Overall, this lab demonstrates both the strength and limitations of classical PCA-based face recognition and basic image enhancement methods.